USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 30 LAB - CFAR Processing
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 4.7: I can apply constant false-alarm rate (CFAR) processing to radar data and evaluate detection performance.

This packet is the L30 Lab, in three parts, and you turn it in once, complete, with your notebook, before Lesson 31. **Part 1, Predict**, is yours to do before the lab: the shaded Predicted cells of each table. **Part 2, Measure**, is your own readings in the white Measured cells of the same tables, from your kit and the notebook, or from the course's recorded data; tick which on each table. **Part 3, Analyze**, is yours to write after the lab.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt. I grade each Part 3 answer on four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before the lab)

1. **Predict.** Fill in every shaded **Predicted** cell of Tables A to F, from the givens below. Lesson 30's Pre-Lab frame names the frame that works each method. Attach your work for one row of each table on your own paper.

    - **The app.** Tables A to C use the Phaser app's Range (CFAR) preset: one chirp of 500 MHz in 0.5 ms each update, a Blackman window, and a profile from 0 to 20 m. The app counts Guard Cells and Ref Cells on each side of the cell under test, uses the reference cells that exist at the ends of the profile, and averages them as power.
    - **Table A** asks whether a target 3 m out is found at the app's opening setting (4 Guard Cells, 16 Ref Cells, 6 dB), at the lesson's window (2, 16, 12.4 dB), and at your own setting with no guard cells. On a typical kit the leakage's first two cells, from 0 to 0.6 m, together hold about 12 dB more power than the target's peak. The target's own peak spills into the cells beside it: about 2 dB down one cell nearer, 7 dB down one cell farther, and 13 dB down two cells nearer. Take your own setting as 4 Ref Cells and 13 dB .
    - **Table B** asks for the farthest range at a bias 3 dB higher, as a fraction of the farthest range at your setting.
    - **Table C** asks how many of 30 updates of an empty scene hold a crossing beyond 1 m, with 3 Guard Cells and 4 Ref Cells at 13, 10, and 7 dB . The profile holds 63 cells beyond 1 m.
    - **Table D** asks for the notebook's 2D bias at 10^{-3} and 10^{-4}, and the false alarms each gives on the empty map. The window has 2 guard and 4 reference cells on each side, in range and in Doppler, and the map tests $64{,}106$ cells.
    - **Table E** asks whether each method finds a weak target four range cells nearer than a strong one, at the same velocity and 8.3 dB below it, with the Hann window on both axes. Every method runs at 11.6 dB .
    - **Table F** asks for P_d at two biases. The notebook sets the hand's strongest cell to 12 dB of SNR and redraws only the noise for each of the 50 maps.

Part 2: Measure (your kit and the notebook, or the recorded data)

Follow the procedure on the L30 page and fill in every white **Measured** cell. A margin is in dB over the threshold, negative below it.

2. **Table A: place the threshold** (Steps 6 to 8). Kit: your stand-in 3 m out. Recorded: `L30_range_person.npz`.
   Kit ▫ Recorded ▫

	Target found?		Target margin	Noise margin
Guard, Ref, Bias	**Predicted**	**Measured**	**Measured**	**Measured**
4, 16, 6 dB				
2, 16, 12.4 dB				
Yours: , ,	–	–		
Yours, 0 guard cells			Threshold rise (dB):	

3. **Table B: walk the line** (Steps 9 to 12). Kit: your screen recordings. Recorded: `L30_walker.npz`.
   Kit ▫ Recorded ▫

Bias	**Nearest (m)**	**Farthest (m)**	**Anything else that crossed**
Your setting			
Your setting $+3$ dB			
Predicted ratio, farthest	–		–
Measured ratio	–		–

4. **Table C: an empty scene** (Steps 13 to 15), 3 Guard Cells and 4 Ref Cells, 30 updates. Kit: your capture in the notebook. Recorded: `L30_range_empty.npz`.
   Kit ▫ Recorded ▫

	Updates with a crossing		
Bias	**Predicted**	**Measured**	**Where they land**
13 dB			
10 dB			
7 dB			

Empty capture:

5. **Table D: notebook task (a)** (Step 17), the empty burst, 2 guard and 4 reference cells each way.
Kit ▫ Recorded ▫

	BIAS_DB		False alarms	
Run	**Predicted**	**Measured**	**Predicted**	**Measured**
Hann, 10^{-3}				
Hann, 10^{-4}				
Rectangular, 10^{-3}	–	–		

6. **Table E: notebook task (b)** (Step 18), the three methods at 11.6 dB .
Kit ▫ Recorded ▫

	Weak target		False-alarm clusters
Method	**Predicted**	**Measured**	**Measured**
`"ca"`			
`"go"`			
`"so"`			

7. **Table F: notebook task (c)** (Steps 19 and 20), P_d over 50 maps, and the detection list.
Kit ▫ Recorded ▫

	P_d	
BIAS_DB_C	**Predicted**	**Measured**
11.6 dB		
Your 10^{-3} bias, dB		

Range (m)	Velocity (m/s)	SNR (dB)	Cells	Target or false alarm

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions.

8. **Where the threshold found you.** Compare the ranges and margins in Tables A and B with your predictions. State what set the nearest and the farthest range at which your threshold found a target, give the evidence from your own tables, and describe a measurement on this kit that would test your explanation.

9. **False alarms against the design.** Compare your counts in Tables C and D with your predictions. Say which agree and which do not, give the cause you think most likely for any that do not, and say how you would check it.

10. **A detector for the L31 Lab.** From your Tables E and F, choose a method and a bias for the L31 Lab's walker records, and argue the choice from your own numbers against at least one alternative you measured.

Documentation: _____